ECE 343 - Chapter 06
Current Density



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Spring 2024

Lesson	Topics	Reading	Notes
21	BVP (Capacitor, Coax), Electric Potential	5.16	
22	Electric Boundary Conditions	5.18-5.22	
23	Capacitance	5.23-5.24	
Chapter 6: Steady Current & Conductivity			
24	Current density, Conductivity, Resistance	Ch. 6	
25	Lab	TBD	
Chapter 7: Magnetostatics			
26	Magnetic Fields, Gauss' Law for Magnetism	7.1-7.3	**GR 2**
27	Ampere's Law	7.4-7.5, 7.9	
28	Ampere's Law Applied	7.6-7.8	
29	Magnetic Boundary Conditions	7.10-7.14	
30	Flex day		
Chapter 8: Time-Varying Fields			
31	Faraday's Law, Displacement Current, Maxwell's Equations	8.3, 8.8, 8.9	
32	Faraday's Law, Displacement Current, Maxwell's Equations	8.3, 8.8, 8.9	
Chapter 9: Plane Waves in Lossless Media			
33	Maxwell's Equations in lossless, source-free region	9.1-9.2	
34	Uniform Plane Waves & Polarization	9.4-9.6	
35	Normal Incidence	Vol 2, 5.6-5.7	
36	Oblique Incidence	Vol 2, 5.6-5.7	
37	Oblique Incidence	Vol 2, 5.6-5.7	
38	Incidence Lab	TBD	
39	Flex day		**GR 3**
40	Wrapup		

Learning Objectives

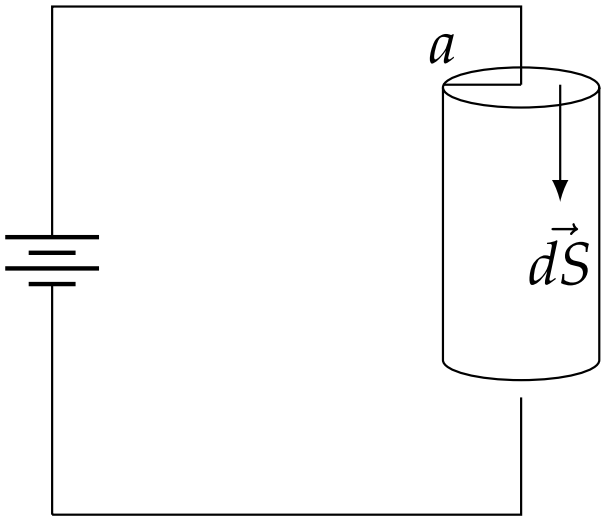
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3.1 Calculate the total current through a surface area given a current distribution.

3.2 Use Ohm's Law for electromagnetics to calculate the relationship between a current distribution and an electric field.

3.3 Articulate and identify what makes a material a good conductor and how conductivity relates to resistance.

Current Distributions



$$I = \int_0^{\rho=a} \int_{\phi=0}^{2\pi} \vec{J} \cdot d\vec{S}$$

$$= 8\mathrm{A/m}^2 \int_0^{\rho=a} \int_{\phi=0}^{2\pi} \rho d\rho d\phi$$

$$= \left(8\mathrm{A/m}^2\right)\left(\frac{a^2}{2}\right)(2\pi)$$

$$= 226\mu\mathrm{A}$$

Conductivity

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Definition

$$\vec{J} = \sigma \vec{E}$$

$$\implies \sigma \longrightarrow \frac{1}{\Omega} = \text{S (Siemens)}$$

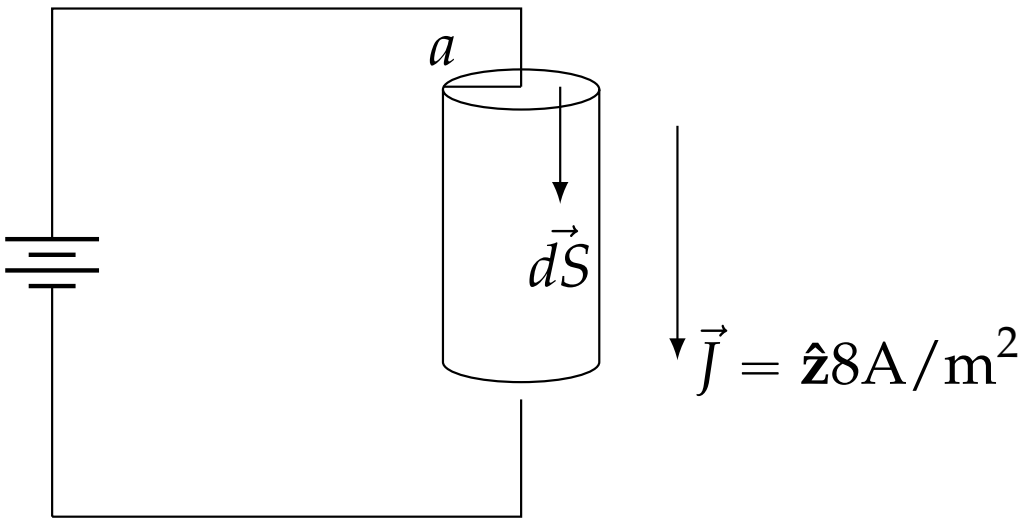
Types of Conductors

1. "Good" conductor $\longrightarrow \sigma > 10^5$ S/m
2. Semiconductor $\longrightarrow \times 10^{-4} < \sigma < 10$ S/m
3. Poor insulators $\longrightarrow 10$ S/m
4. Good insulators $\longrightarrow \sigma < \times 10^{-10}$ S/m
5. Vacuum $\longrightarrow 0$

Conductivity for Common Materials

Material	σ (S/m)
Copper	5.8×10^7
Gold	4.4×10^7
Aluminum	3.7×10^7
Iron	1.0×10^7
Platinum	0.9×10^7
Carbon	1.3×10^5
Silicon	4.4×10^{-4}
Free space	≈ 0
Distilled water	6×10^{-6}
Tap water	$5 - 50 \times 10^{-3}$
Seawater	5
Soil	1×10^{-4}

Resistance of a Wire



$$V = -\int \vec{E}\cdot d\vec{\ell}$$

$$= -\int (-\hat{\mathbf{z}}E_z)\cdot\hat{\mathbf{z}}dz$$

$$= E_z\ell$$

$$\Longrightarrow \vec{J} = \sigma\vec{E} = \hat{\mathbf{z}}\sigma E_z$$

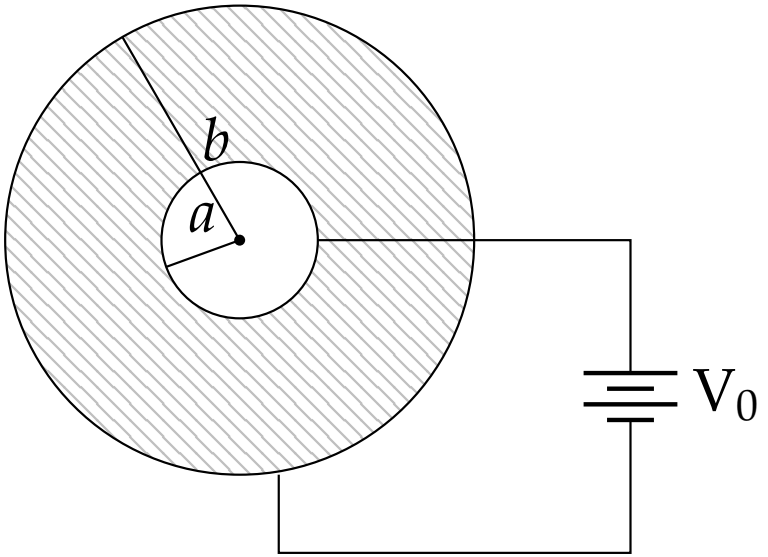
$$\Longrightarrow I = \int_{\rho=0}^{a}\int_{\phi=0}^{2\pi} \hat{\mathbf{z}}\sigma E_z\cdot\hat{\mathbf{z}}\rho d\rho d\phi$$

$$= \sigma E_z \pi a^2$$

$$\Longrightarrow R = \frac{V}{I} = \frac{E_z\ell}{\sigma E_z\pi a^2} = \frac{\ell}{\sigma A}$$

Resistance of a Coaxial Cable

Find the resistance between the inner and outer conductor of a coaxial cable of length (in z-dimension) ℓ.



$$\vec{E} = \hat{\boldsymbol{\rho}} \frac{Q}{2\pi\varepsilon_0 \rho L}$$

$$V = -\int_b^a \frac{Q}{2\pi\varepsilon_0 \rho L}$$

$$= \frac{Q}{2\pi\varepsilon_0 \rho L} \ln\left(\frac{b}{a}\right)$$

$$I = \int \vec{J} \cdot d\vec{S} = \int \sigma \vec{E} \cdot d\vec{S}$$

$$= \int \sigma \hat{\boldsymbol{\rho}} \frac{Q}{2\pi\varepsilon_0 \rho L} \cdot \hat{\boldsymbol{\rho}} \rho d\phi dz$$

$$= \frac{\sigma Q}{2\pi\varepsilon_0 L} \int_0^{2\pi} d\phi \int_0^L dz = \frac{\sigma Q}{\varepsilon_0}$$

$$R = \frac{V}{I} = \left[\frac{Q}{2\pi\varepsilon_0 \rho L} \ln\left(\frac{b}{a}\right)\right]\left[\frac{\varepsilon_0}{\sigma Q}\right]$$

$$= \frac{1}{2\pi\sigma L} \ln\left(\frac{b}{a}\right)$$

$$\Longrightarrow G' = \frac{G}{L} = \frac{1}{RL} = \frac{2\pi\sigma}{\ln\left(\frac{b}{a}\right)}$$